1) a)

The Restoring Force acts **towards** O, so it acts in the opposite direction to displacement:

Restoring Force $= -10x$ N

1) b)

The Resistance Force always opposes motion, so if the velocity $v = \frac{\mathrm{d}x}{\mathrm{d}t}$ then:

Resistance Force $= -6v = -6\frac{\mathrm{d}x}{\mathrm{d}t}$ N

This makes sense as Resistance Force < 0 when moving to the right (as $\frac{\mathrm{d}x}{\mathrm{d}t}$ will be positive), and Resistance Force > 0 when moving to the left (as $\frac{\mathrm{d}x}{\mathrm{d}t}$ will be negative), and so it always opposes motion.

$\therefore$ Resistance Force $= -6\frac{\mathrm{d}x}{\mathrm{d}t}$ N

1) c)

Resultant Force $F = ma$

$m = 2$ and $a = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$\Rightarrow -10x - 6\frac{\mathrm{d}x}{\mathrm{d}t} = 2\frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$\Rightarrow 2\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 6\frac{\mathrm{d}x}{\mathrm{d}t} + 10x = 0$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 3\frac{\mathrm{d}x}{\mathrm{d}t} + 5x = 0$

2)

The positive direction is away from A, so the force of magnitude $4x$ N acting towards A will be $-4x$ N.

The resistance force will act against the direction of motion, so:

$-2v = -2\frac{\mathrm{d}x}{\mathrm{d}t}$

So:

Resultant Force $F = ma$

$m = 0.5$ and $a = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$\Rightarrow -4x - 2\frac{\mathrm{d}x}{\mathrm{d}t} = 0.5\frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$\Rightarrow 0.5\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 2\frac{\mathrm{d}x}{\mathrm{d}t} + 4x = 0$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 4\frac{\mathrm{d}x}{\mathrm{d}t} + 8x = 0$

3)

The positive direction is away from O.

There is a force acting on P directed towards O:

Force $= -kx$ N

The resistance force will act against the direction of motion:

Resistance Force $= -\lambda v = -\lambda\frac{\mathrm{d}x}{\mathrm{d}t}$ N

So:

Resultant Force $F = ma$

$a = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$\Rightarrow -kx - \lambda\frac{\mathrm{d}x}{\mathrm{d}t} = m\frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$\Rightarrow m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \lambda\frac{\mathrm{d}x}{\mathrm{d}t} + kx = 0$

4)

The restoring force will act towards the origin, so in the negative x-direction:

Restoring Force $= -12x$ N

The resistance force will act against the direction of motion:

Resistance Force $= -9v = -9\frac{\mathrm{d}x}{\mathrm{d}t}$ N

The driving force acts in the positive x-direction:

Driving Force $= 15\cos t$ N

So:

Resultant Force $F = ma$

$m = 3$ and $a = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$\Rightarrow -12x - 9\frac{\mathrm{d}x}{\mathrm{d}t} + 15\cos t = 3\frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$\Rightarrow 3\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 9\frac{\mathrm{d}x}{\mathrm{d}t} + 12x = 15\cos t$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 3\frac{\mathrm{d}x}{\mathrm{d}t} + 4x = 5\cos t$

5) a)

The restoring force acts towards the origin, so in the negative x-direction:

Restoring Force $= -8x$ N

The resistance force will act against the direction of motion:

Resistance Force $= -4v = -4\frac{\mathrm{d}x}{\mathrm{d}t}$ N

So:

Resultant Force $F = ma$

$m = 1$ and $a = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$\Rightarrow -8x - 4\frac{\mathrm{d}x}{\mathrm{d}t} = 1 \times \frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 4\frac{\mathrm{d}x}{\mathrm{d}t} + 8x = 0$

5) b)

Auxiliary Equation:

$m^2 + 4m + 8 = 0$

$\Rightarrow (m+2)^2 - 4 + 8 = 0$

$\Rightarrow (m+2)^2 = -4$

$\Rightarrow m = -2 \pm 2\mathrm{i}$

General Solution:

$x = \mathrm{e}^{-2t}(A\cos 2t + B\sin 2t)$

Substitute $x = 2$ and $t = 0$:

$2 = \mathrm{e}^{0}(A\cos 0 + B\sin 0)$

$\Rightarrow 2 = A$

$\therefore x = \mathrm{e}^{-2t}(2\cos 2t + B\sin 2t)$

$\frac{\mathrm{d}x}{\mathrm{d}t} = \mathrm{e}^{-2t}(-4\sin 2t + 2B\cos 2t) - 2\mathrm{e}^{-2t}(2\cos 2t + B\sin 2t)$

Substitute $\frac{\mathrm{d}x}{\mathrm{d}t} = 0$ and $t = 0$:

$0 = \mathrm{e}^{0}(-4\sin 0 + 2B\cos 0) - 2\mathrm{e}^{0}(2\cos 0 + B\sin 0)$

$\Rightarrow 0 = 2B - 4 \Rightarrow B = 2$

$\therefore x = \mathrm{e}^{-2t}(2\cos 2t + 2\sin 2t)$

$\Rightarrow x = 2\mathrm{e}^{-2t}(\cos 2t + \sin 2t)$

6)

The restoring force acts towards the origin, so in the negative x-direction:

Restoring Force $= -6x$ N

The resistive force will act against the direction of motion:

Resistance Force $= -8v = -8\frac{\mathrm{d}x}{\mathrm{d}t}$ N

So:

Resultant Force $F = ma$

$m = 2$ and $a = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$\Rightarrow -6x - 8\frac{\mathrm{d}x}{\mathrm{d}t} = 2\frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$\Rightarrow 2\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 8\frac{\mathrm{d}x}{\mathrm{d}t} + 6x = 0$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 4\frac{\mathrm{d}x}{\mathrm{d}t} + 3x = 0$

Auxiliary Equation:

$m^2 + 4m + 3 = 0$

$\Rightarrow (m+3)(m+1) = 0$

$\Rightarrow m = -3$ or $m = -1$

General Solution:

$x = A\mathrm{e}^{-3t} + B\mathrm{e}^{-t}$

Substitute $x = 3$ and $t = 0$:

$3 = A\mathrm{e}^0 + B\mathrm{e}^0$

$\Rightarrow A + B = 3$ (1)

$\frac{\mathrm{d}x}{\mathrm{d}t} = -3A\mathrm{e}^{-3t} - B\mathrm{e}^{-t}$

Substitute $\frac{\mathrm{d}x}{\mathrm{d}t} = -1$ and $t = 0$:

$-1 = -3A\mathrm{e}^0 - B\mathrm{e}^0$

$\Rightarrow -3A - B = -1 \Rightarrow 3A + B = 1$ (2)

Solve (1) & (2):

$A = -1$ and $B = 4$

$\therefore x = -\mathrm{e}^{-3t} + 4\mathrm{e}^{-t}$

7) a)

The force of magnitude $5x$ N acts in the direction PO:

Force $= -5x$ N

The resistance force will act against the direction of motion:

Resistance Force $= -2v = -2\frac{\mathrm{d}x}{\mathrm{d}t}$ N

So:

Resultant Force $F = ma$

$m = 0.8$ and $a = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$\Rightarrow -5x - 2\frac{\mathrm{d}x}{\mathrm{d}t} = 0.8\frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$\Rightarrow 0.8\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 2\frac{\mathrm{d}x}{\mathrm{d}t} + 5x = 0$

$\Rightarrow 4\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 10\frac{\mathrm{d}x}{\mathrm{d}t} + 25x = 0$

7) b)

Auxiliary Equation:

$4m^2 + 10m + 25 = 0$

$\Rightarrow 4\left(m^2 + \frac{5}{2}m\right) + 25 = 0$

$\Rightarrow 4\left(\left(m + \frac{5}{4}\right)^2 - \frac{25}{16}\right) + 25 = 0$

$\Rightarrow 4\left(m + \frac{5}{4}\right)^2 - \frac{25}{4} + 25 = 0$

$\Rightarrow \left(m + \frac{5}{4}\right)^2 = -\frac{75}{16}$

$\Rightarrow m = -\frac{5}{4} \pm \frac{5\sqrt{3}}{4}\mathrm{i}$

General Solution:

$x = \mathrm{e}^{-\frac{5}{4}t}\left(A\cos\left(\frac{5\sqrt{3}}{4}t\right) + B\sin\left(\frac{5\sqrt{3}}{4}t\right)\right)$

Substitute $x = 1.2$ and $t = 0$:

$1.2 = \mathrm{e}^0(A\cos 0 + B\sin 0)$

$\Rightarrow 1.2 = A$

$\therefore x = \mathrm{e}^{-\frac{5}{4}t}\left(1.2\cos\left(\frac{5\sqrt{3}}{4}t\right) + B\sin\left(\frac{5\sqrt{3}}{4}t\right)\right)$

$\frac{\mathrm{d}x}{\mathrm{d}t} = \mathrm{e}^{-\frac{5}{4}t}\left(-\frac{3\sqrt{3}}{2}\sin\left(\frac{5\sqrt{3}}{4}t\right) + \frac{5\sqrt{3}}{4}B\cos\left(\frac{5\sqrt{3}}{4}t\right)\right) - \frac{5}{4}\mathrm{e}^{-\frac{5}{4}t}\left(1.2\cos\left(\frac{5\sqrt{3}}{4}t\right) + B\sin\left(\frac{5\sqrt{3}}{4}t\right)\right)$

Substitute $\frac{\mathrm{d}x}{\mathrm{d}t} = 3$ and $t = 0$:

$3 = \mathrm{e}^0\left(-\frac{3\sqrt{3}}{2}\sin 0 + \frac{5\sqrt{3}}{4}B\cos 0\right) - \frac{5}{4}\mathrm{e}^0(1.2\cos 0 + B\sin 0)$

$\Rightarrow 3 = \frac{5\sqrt{3}}{4}B - \frac{3}{2} \Rightarrow B = \frac{6\sqrt{3}}{5}$

$\therefore x = \mathrm{e}^{-\frac{5}{4}t}\left(\frac{6}{5}\cos\left(\frac{5\sqrt{3}}{4}t\right) + \frac{6\sqrt{3}}{5}\sin\left(\frac{5\sqrt{3}}{4}t\right)\right)$

$x|_{t=1} = 0.301$ m (3 SF)

8) a)

The restoring force acts towards the origin, so in the negative x-direction:

Restoring Force $= -7.5x$ N

The resistance force will act against the direction of motion:

Resistance Force $= -6v = -6\frac{\mathrm{d}x}{\mathrm{d}t}$ N

The driving force acts in the positive x-direction:

Driving Force $= 9\sin 2t$ N

So:

Resultant Force $F = ma$

$m = 1.5$ and $a = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$\Rightarrow -7.5x - 6\frac{\mathrm{d}x}{\mathrm{d}t} + 9\sin 2t = 1.5\frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$\Rightarrow 1.5\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 6\frac{\mathrm{d}x}{\mathrm{d}t} + 7.5x = 9\sin 2t$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 4\frac{\mathrm{d}x}{\mathrm{d}t} + 5x = 6\sin 2t$

8) b)

Homogeneous Equation:

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 4\frac{\mathrm{d}x}{\mathrm{d}t} + 5x = 0$

Auxiliary Equation:

$m^2 + 4m + 5 = 0$

$\Rightarrow (m+2)^2 - 4 + 5 = 0$

$\Rightarrow (m+2)^2 = -1$

$\Rightarrow m = -2 \pm \mathrm{i}$

Complementary Function:

$x = \mathrm{e}^{-2t}(A\cos t + B\sin t)$

Try this Particular Integral:

$x = p\cos 2t + q\sin 2t$

$\frac{\mathrm{d}x}{\mathrm{d}t} = -2p\sin 2t + 2q\cos 2t$

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -4p\cos 2t - 4q\sin 2t$

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 4\frac{\mathrm{d}x}{\mathrm{d}t} + 5x = 6\sin 2t$

$\Rightarrow (-4p\cos 2t - 4q\sin 2t) + 4(-2p\sin 2t + 2q\cos 2t) + 5(p\cos 2t + q\sin 2t) = 6\sin 2t$

$\Rightarrow (-4p + 8q + 5p)\cos 2t + (-4q - 8p + 5q)\sin 2t = 6\sin 2t$

$\Rightarrow (p + 8q)\cos 2t + (-8p + q)\sin 2t = 6\sin 2t$

Compare Coefficients:

$p + 8q = 0$ (1)

$-8p + q = 6$ (2)

Solve (1) & (2):

$p = -\frac{48}{65}$ and $q = \frac{6}{65}$

Particular Integral:

$x = -\frac{48}{65}\cos 2t + \frac{6}{65}\sin 2t$

General Solution:

$x = \mathrm{e}^{-2t}(A\cos t + B\sin t) - \frac{48}{65}\cos 2t + \frac{6}{65}\sin 2t$

Substitute $x = 4$ and $t = 0$:

$4 = \mathrm{e}^{0}(A\cos 0 + B\sin 0) - \frac{48}{65}\cos 0 + \frac{6}{65}\sin 0$

$\Rightarrow 4 = A - \frac{48}{65} \Rightarrow A = \frac{308}{65}$

$\therefore x = \mathrm{e}^{-2t}\left(\frac{308}{65}\cos t + B\sin t\right) - \frac{48}{65}\cos 2t + \frac{6}{65}\sin 2t$

$\frac{\mathrm{d}x}{\mathrm{d}t} = \mathrm{e}^{-2t}\left(-\frac{308}{65}\sin t + B\cos t\right) - 2\mathrm{e}^{-2t}\left(\frac{308}{65}\cos t + B\sin t\right) + \frac{96}{65}\sin 2t + \frac{12}{65}\cos 2t$

Substitute $\frac{\mathrm{d}x}{\mathrm{d}t} = 1$ and $t = 0$:

$1 = \mathrm{e}^{0}\left(-\frac{308}{65}\sin 0 + B\cos 0\right) - 2\mathrm{e}^{0}\left(\frac{308}{65}\cos 0 + B\sin 0\right) + \frac{96}{65}\sin 0 + \frac{12}{65}\cos 0$

$\Rightarrow 1 = B - \frac{616}{65} + \frac{12}{65}$

$\Rightarrow B = \frac{669}{65}$

Particular Solution:

$x = \mathrm{e}^{-2t}\left(\frac{308}{65}\cos t + \frac{669}{65}\sin t\right) - \frac{48}{65}\cos 2t + \frac{6}{65}\sin 2t$

8) c)

As $t \to \infty$, $\mathrm{e}^{-2t} \Rightarrow 0$, so $x \to -\frac{48}{65}\cos 2t + \frac{6}{65}\sin 2t$

So the motion approaches steady oscillation (simple harmonic motion).

9) a)

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 6\frac{\mathrm{d}x}{\mathrm{d}t} + 25x = 0$

Auxiliary Equation:

$m^2 + 6m + 25 = 0$

$\Rightarrow (m+3)^2 - 9 + 25 = 0$

$\Rightarrow (m+3)^2 = -16$

$\Rightarrow m = -3 \pm 4\mathrm{i}$

Complementary Function:

$x = \mathrm{e}^{-3t}(A\cos 4t + B\sin 4t)$

(Taking downwards as positive)

Substitute $x = 0.08$ and $t = 0$:

$0.08 = \mathrm{e}^0(A\cos 0 + B\sin 0)$

$\Rightarrow 0.08 = A$

$\therefore x = \mathrm{e}^{-3t}(0.08\cos 4t + B\sin 4t)$

$\frac{\mathrm{d}x}{\mathrm{d}t} = \mathrm{e}^{-3t}(-0.32\sin 4t + 4B\cos 4t) - 3\mathrm{e}^{-3t}(0.08\cos 4t + B\sin 4t)$

Substitute $\frac{\mathrm{d}x}{\mathrm{d}t} = 0$ and $t = 0$:

$0 = \mathrm{e}^0(-0.32\sin 0 + 4B\cos 0) - 3\mathrm{e}^0(0.08\cos 0 + B\sin 0)$

$\Rightarrow 0 = 4B - 0.24 \Rightarrow B = 0.06$

$\therefore x = \mathrm{e}^{-3t}(0.08\cos 4t + 0.06\sin 4t)$

9) b)

The roots of the auxiliary equation are $-3 \pm 4\mathrm{i}$, so the solution contains sine and cosine terms. Therefore the system undergoes damped oscillations (the real part of the solutions are negative).

9) c)

$\frac{\mathrm{d}x}{\mathrm{d}t} = \mathrm{e}^{-3t}(-0.32\sin 4t + 0.24\cos 4t) - 3\mathrm{e}^{-3t}(0.08\cos 4t + 0.06\sin 4t)$

The maximum displacement occurs when $\frac{\mathrm{d}x}{\mathrm{d}t} = 0$

$0 = \mathrm{e}^{-3t}(-0.32\sin 4t + 0.24\cos 4t) - 3\mathrm{e}^{-3t}(0.08\cos 4t + 0.06\sin 4t)$

$0 = -0.32\sin 4t + 0.24\cos 4t - 3(0.08\cos 4t + 0.06\sin 4t)$

$0 = -0.32\sin 4t + 0.24\cos 4t - 0.24\cos 4t - 0.18\sin 4t$

$0 = -0.5\sin 4t$

$\Rightarrow \sin 4t = 0$

$\Rightarrow 4t = \pi$

$\Rightarrow t = \frac{\pi}{4}$ seconds

10) a)

$\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} + 2\frac{\mathrm{d}\theta}{\mathrm{d}t} + 10\theta = 0$

Auxiliary Equation:

$m^2 + 2m + 10 = 0$

$\Rightarrow (m+1)^2 - 1 + 10 = 0$

$\Rightarrow (m+1)^2 = -9$

$\Rightarrow m = -1 \pm 3\mathrm{i}$

Complementary Function:

$x = \mathrm{e}^{-t}(A\cos 3t + B\sin 3t)$

Substitute $\theta = 0.2$ and $t = 0$:

$0.2 = \mathrm{e}^{0}(A\cos 0 + B\sin 0)$

$\Rightarrow 0.2 = A$

$\therefore x = \mathrm{e}^{-t}(0.2\cos 3t + B\sin 3t)$

$\frac{\mathrm{d}\theta}{\mathrm{d}t} = \mathrm{e}^{-t}(-0.6\sin 3t + 3B\cos 3t) - \mathrm{e}^{-t}(0.2\cos 3t + B\sin 3t)$

Substitute $\frac{\mathrm{d}\theta}{\mathrm{d}t} = 0$ and $t = 0$:

$0 = \mathrm{e}^{0}(-0.6\sin 0 + 3B\cos 0) - \mathrm{e}^{0}(0.2\cos 0 + B\sin 0)$

$\Rightarrow 0 = 3B - 0.2 \Rightarrow B = \frac{1}{15}$

$\therefore \theta = \mathrm{e}^{-3t}\left(\frac{1}{5}\cos 4t + \frac{1}{15}\sin 4t\right)$

10) b)

As the auxiliary equation has two complex conjugate roots, the system undergoes Light Damping (underdamping).

10) c)

As $t \to \infty$, $\mathrm{e}^{-t} \to 0$ so $\theta \to 0$

So the panel oscillates with decreasing amplitude and tends towards rest at equilibrium.

11) a)

The restoring force acts towards the equilibrium position, so in the negative x-direction:

Restoring Force $= -18x$ N

The resistance force will act against the direction of motion:

Resistance Force $= -\lambda v = -\lambda \frac{\mathrm{d}x}{\mathrm{d}t}$ N

So:

Resultant Force $F = ma$

$a = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$\Rightarrow -18x - \lambda \frac{\mathrm{d}x}{\mathrm{d}t} = m \frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$\Rightarrow m \frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \lambda \frac{\mathrm{d}x}{\mathrm{d}t} + 18x = 0$

11) b)

Auxiliary Equation:

$mr^2 + \lambda r + 18 = 0$

11) c) i)

$m = 2$ and $\lambda = 8$:

$2r^2 + 8r + 18 = 0$

$\Rightarrow r^2 + 4r + 9 = 0$

$\Rightarrow (r + 2)^2 - 4 + 9 = 0$

$\Rightarrow (r + 2)^2 = -5$

$\Rightarrow r = -2 \pm \sqrt{5}\mathrm{i}$

General Solution:

$x = \mathrm{e}^{-2t}\left(A\cos\left(\sqrt{5}t\right) + B\sin\left(\sqrt{5}t\right)\right)$

11) c) ii)

The roots of the auxiliary equation are $-2 \pm \sqrt{5}\mathrm{i}$, so the solution contains sine and cosine terms. Therefore the system undergoes damped oscillations (the real part of the solutions are negative).

11) d)

$m = 2$:

$2r^2 + \lambda r + 18 = 0$

Critical Damping $\Rightarrow b^2 - 4ac = 0$

$a = 2$

$b = \lambda$

$c = 18$

$\therefore \lambda^2 - 4 \times 2 \times 18 = 0$

$\Rightarrow \lambda^2 - 144 = 0$

$\Rightarrow \lambda = \pm 12$

For critical damping, $\lambda > 0$ so $\lambda = 12$

11) e)

If $m = 3$ and $\lambda = 12$:

$3r^2 + 12r + 18 = 0$

$\Rightarrow r^2 + 4r + 6 = 0$

$\Rightarrow (r+2)^2 - 4 + 6 = 0$

$\Rightarrow (r+2)^2 = -2$

$\Rightarrow r = -2 \pm \sqrt{2}\mathrm{i}$

As the solutions are complex, and the real part is negative, the motion of the mass will be Light Damping (underdamping), where the system undergoes damped oscillations

12)

$2\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 7k\frac{\mathrm{d}x}{\mathrm{d}t} + 6k^2x = 4kU$

Homogeneous Equation:

$2\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 7k\frac{\mathrm{d}x}{\mathrm{d}t} + 6k^2x = 0$

Auxiliary Equation:

$2m^2 + 7km + 6k^2 = 0$

$\Rightarrow 2\left(m^2 + \frac{7}{2}km\right) + 6k^2 = 0$

$\Rightarrow 2\left(\left(m + \frac{7}{4}k\right)^2 - \frac{49}{16}k^2\right) + 6k^2 = 0$

$\Rightarrow 2\left(m + \frac{7}{4}k\right)^2 - \frac{49}{8}k^2 + 6k^2 = 0$

$\Rightarrow 2\left(m + \frac{7}{4}k\right)^2 - \frac{1}{8}k^2 = 0$

$\Rightarrow \left(m + \frac{7}{4}k\right)^2 = \frac{1}{16}k^2$

$\Rightarrow m = -\frac{7}{4}k \pm \frac{1}{4}k$

$\Rightarrow m = -\frac{3}{2}k$ or $m = -2k$

Complementary Function:

$x = A\mathrm{e}^{-\frac{3}{2}kt} + B\mathrm{e}^{-2kt}$

Try this Particular Integral:

$x = c$

$\frac{\mathrm{d}x}{\mathrm{d}t} = 0$

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = 0$

$2\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 7k\frac{\mathrm{d}x}{\mathrm{d}t} + 6k^2x = 4kU$

$\Rightarrow 2(0) + 7k(0) + 6k^2(c) = 4kU$

$\Rightarrow c = \frac{4kU}{6k^2} = \frac{2U}{3k}$

Particular Integral:

$x = \frac{2U}{3k}$

General Solution:

$x = A\mathrm{e}^{-\frac{3}{2}kt} + B\mathrm{e}^{-2kt} + \frac{2U}{3k}$

Substitute $x = 0$ and $t = 0$:

$0 = A\mathrm{e}^{0} + B\mathrm{e}^{0} + \frac{2U}{3k}$

$\Rightarrow A + B = -\frac{2U}{3k}$ (1)

$\frac{\mathrm{d}x}{\mathrm{d}t} = -\frac{3}{2}Ak\mathrm{e}^{-\frac{3}{2}kt} - 2Bk\mathrm{e}^{-2kt}$

Substitute $\frac{\mathrm{d}x}{\mathrm{d}t} = 0$ and $t = 0$:

$0 = -\frac{3}{2}Ak\mathrm{e}^{0} - 2Bk\mathrm{e}^{0}$

$\Rightarrow \frac{3}{2}A + 2B = 0$

$\Rightarrow A = -\frac{4}{3}B$ (2)

Substitute (2) into (1):

$-\frac{4}{3}B + B = -\frac{2U}{3k} \Rightarrow -\frac{1}{3}B = -\frac{2U}{3k} \Rightarrow B = \frac{2U}{k}$

$\therefore A = -\frac{4}{3} \times \frac{2U}{k} = -\frac{8U}{3k}$

$\therefore x = -\frac{8U}{3k}\mathrm{e}^{-\frac{3}{2}kt} + \frac{2U}{k}\mathrm{e}^{-2kt} + \frac{2U}{3k}$

13) a)

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 4\frac{\mathrm{d}x}{\mathrm{d}t} + 4x = 10\mathrm{e}^{-2t}$

Homogeneous Equation:

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 4\frac{\mathrm{d}x}{\mathrm{d}t} + 4x = 0$

Auxiliary Equation:

$m^2 + 4m + 4 = 0$

$\Rightarrow (m+2)^2 = 0$

$\Rightarrow m = -2$

Complementary Function:

$x = At\mathrm{e}^{-2t} + B\mathrm{e}^{-2t}$

Try this Particular Integral:

$x = pt^2\mathrm{e}^{-2t}$

$\frac{\mathrm{d}x}{\mathrm{d}t} = -2pt^2\mathrm{e}^{-2t} + 2pt\mathrm{e}^{-2t}$

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = 4pt^2\mathrm{e}^{-2t} - 4pt\mathrm{e}^{-2t} - 4pt\mathrm{e}^{-2t} + 2p\mathrm{e}^{-2t}$

$= 4pt^2\mathrm{e}^{-2t} - 8pt\mathrm{e}^{-2t} + 2p\mathrm{e}^{-2t}$

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 4\frac{\mathrm{d}x}{\mathrm{d}t} + 4x = 10\mathrm{e}^{-2t}$

$\Rightarrow (4pt^2\mathrm{e}^{-2t} - 8pt\mathrm{e}^{-2t} + 2p\mathrm{e}^{-2t}) + 4(-2pt^2\mathrm{e}^{-2t} + 2pt\mathrm{e}^{-2t}) + 4(pt^2\mathrm{e}^{-2t}) = 10\mathrm{e}^{-2t}$

$\Rightarrow (4p - 8p + 4p)t^2\mathrm{e}^{-2t} + (8p + 8p)t\mathrm{e}^{-2t} + (2p)\mathrm{e}^{-2t} = 10\mathrm{e}^{-2t}$

$\Rightarrow (2p)\mathrm{e}^{-2t} = 10\mathrm{e}^{-2t}$

$\Rightarrow 2p = 10$

$\Rightarrow p = 5$

Particular Integral:

$x = 5t^2\mathrm{e}^{-2t}$

General Solution:

$x = At\mathrm{e}^{-2t} + B\mathrm{e}^{-2t} + 5t^2\mathrm{e}^{-2t}$

Substitute $x = 18$ and $t = 0$:

$18 = A \times 0 \times \mathrm{e}^0 + B\mathrm{e}^0 + 5 \times 0^2 \times \mathrm{e}^0$

$\Rightarrow B = 18$

$\therefore x = At\mathrm{e}^{-2t} + 18\mathrm{e}^{-2t} + 5t^2\mathrm{e}^{-2t}$

$\frac{\mathrm{d}x}{\mathrm{d}t} = -2At\mathrm{e}^{-2t} + A\mathrm{e}^{-2t} - 36\mathrm{e}^{-2t} - 10t^2\mathrm{e}^{-2t} + 10t\mathrm{e}^{-2t}$

Substitute $\frac{\mathrm{d}x}{\mathrm{d}t} = 8$ and $t = 0$:

$8 = -2A \times 0 \times \mathrm{e}^0 + A\mathrm{e}^0 - 36\mathrm{e}^0 - 10 \times 0^2 \times \mathrm{e}^0 + 10 \times 0 \times \mathrm{e}^0$

$\Rightarrow 8 = A - 36$

$\Rightarrow A = 44$
$\therefore x = 44te^{-2t} + 18e^{-2t} + 5t^2e^{-2t}$
$\Rightarrow x = (5t^2 + 44t + 18)e^{-2t}$

13) b)

$\frac{dx}{dt} = (10t + 44)e^{-2t} - 2(5t^2 + 44t + 18)e^{-2t}$
Maximum displacement occurs when $\frac{dx}{dt} = 0$
$(10t + 44)e^{-2t} - 2(5t^2 + 44t + 18)e^{-2t} = 0$
$\Rightarrow (10t + 44)e^{-5t} + (-10t^2 - 88t - 36)e^{-2t} = 0$
$\Rightarrow (-10t^2 - 78t + 8)e^{-5t} = 0$
$\Rightarrow -10t^2 - 78t + 8 = 0$
$\Rightarrow 5t^2 + 39t - 4 = 0$
$\Rightarrow t = \frac{-39 \pm \sqrt{1601}}{10} = -7.901249 \ldots$ or $0.101249 \ldots$
$t \geq 0$ so $t \neq -7.901249 \ldots$
$x|_{t=0.101249\ldots} = 18.380556 \ldots$
$= 18.4$ cm to the nearest mm

13) c)

The mass is at equilibrium when $x = 0$
$\therefore 0 = (5t^2 + 44t + 18)e^{-2t}$
$e^{-2t} \neq 0$ for all $t \geq 0$
So we must have that $5t^2 + 44t + 18 = 0$
However, for $t \geq 0$ we have that $5t^2 \geq 0$, $44t \geq 0$ and $18 > 0$
So $5t^2 + 44t + 18 > 0$
Hence $x > 0$ for all $t \geq 0$ and so the model predicts that the mass never returns to its equilibrium position.

14) a)

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 6\frac{\mathrm{d}x}{\mathrm{d}t} + 13x = 10\cos 2t$

Homogeneous Equation:

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 6\frac{\mathrm{d}x}{\mathrm{d}t} + 13x = 0$

Auxiliary Equation:

$m^2 + 6m + 13 = 0$

$\Rightarrow (m+3)^2 - 9 + 13 = 0$

$\Rightarrow (m+3)^2 = -4$

$\Rightarrow m = -3 \pm 2\mathrm{i}$

Complementary Function:

$x = \mathrm{e}^{-3t}(A\cos 2t + B\sin 2t)$

Try this Particular Integral:

$x = p\cos 2t + q\sin 2t$

$\frac{\mathrm{d}x}{\mathrm{d}t} = -2p\sin 2t + 2q\cos 2t$

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -4p\cos 2t - 4q\sin 2t$

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 6\frac{\mathrm{d}x}{\mathrm{d}t} + 13x = 10\cos 2t$

$\Rightarrow (-4p\cos 2t - 4q\sin 2t) + 6(-2p\sin 2t + 2q\cos 2t) + 13(p\cos 2t + q\sin 2t) = 10\cos 2t$

$\Rightarrow (-4p + 12q + 13p)\cos 2t + (-4q - 12p + 13q)\sin 2t = 10\cos 2t$

$\Rightarrow (9p + 12q)\cos 2t + (-12p + 9q)\sin 2t = 10\cos 2t$

Compare Coefficients:

$9p + 12q = 10$ (1)

$-12p + 9q = 0$ (2)

Solve (1) & (2):

$p = \frac{2}{5}$ and $q = \frac{8}{15}$

Particular Integral:

$x = \frac{2}{5}\cos 2t + \frac{8}{15}\sin 2t$

General Solution:

$x = \mathrm{e}^{-3t}(A\cos 2t + B\sin 2t) + \frac{2}{5}\cos 2t + \frac{8}{15}\sin 2t$

14) b)

The steady-state solution is the part that remains as t becomes large, so

$x = \frac{2}{5}\cos 2t + \frac{8}{15}\sin 2t$

14) c)

Amplitude $= \sqrt{\left(\frac{2}{5}\right)^2 + \left(\frac{8}{15}\right)^2} = \sqrt{\frac{4}{9}} = \frac{2}{3}$ m

15) a)

$\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} + 8\frac{\mathrm{d}\theta}{\mathrm{d}t} + 32\theta = 80\cos 4t$

Homogeneous Equation:

$\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} + 8\frac{\mathrm{d}\theta}{\mathrm{d}t} + 32\theta = 0$

Auxiliary Equation:

$m^2 + 8m + 32 = 0$

$\Rightarrow (m+4)^2 - 16 + 32 = 0$

$\Rightarrow (m+4)^2 = -16$

$\Rightarrow m = -4 \pm 4\mathrm{i}$

Complementary Function:

$\theta = \mathrm{e}^{-4t}(A\cos 4t + B\sin 4t)$

Try this Particular Integral:

$\theta = p\cos 4t + q\sin 4t$

$\frac{\mathrm{d}\theta}{\mathrm{d}t} = -4p\sin 4t + 4q\cos 4t$

$\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} = -16p\cos 4t - 16q\sin 4t$

$\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} + 8\frac{\mathrm{d}\theta}{\mathrm{d}t} + 32\theta = 80\cos 4t$

$\Rightarrow (-16p\cos 4t - 16q\sin 4t) + 8(-4p\sin 4t + 4q\cos 4t) + 32(p\cos 4t + q\sin 4t) = 80\cos 4t$

$\Rightarrow (-16p + 32q + 32p)\cos 4t + (-16q - 32p + 32q)\sin 4t = 80\cos 4t$

$\Rightarrow (16p + 32q)\cos 4t + (-32p + 16q)\sin 4t = 80\cos 4t$

Compare Coefficients:

$16p + 32q = 80$ (1)

$-32p + 16q = 0$ (2)

Solve (1) & (2):

$p = 1$ and $q = 2$

Particular Integral:

$\theta = \cos 4t + 2\sin 4t$

General Solution:

$\theta = \mathrm{e}^{-4t}(A\cos 4t + B\sin 4t) + \cos 4t + 2\sin 4t$

15) b)

Substitute $\theta = 0$ and $t = 0$:

$0 = \mathrm{e}^{0}(A\cos 0 + B\sin 0) + \cos 0 + 2\sin 0$

$\Rightarrow 0 = A + 1$

$\Rightarrow A = -1$

$\therefore \theta = \mathrm{e}^{-4t}(-\cos 4t + B\sin 4t) + \cos 4t + 2\sin 4t$

$\frac{\mathrm{d}\theta}{\mathrm{d}t} = \mathrm{e}^{-4t}(4\sin 4t + 4B\cos 4t) - 4\mathrm{e}^{-4t}(-\cos 4t + B\sin 4t) - 4\sin 4t + 8\cos 4t$

Substitute $\frac{\mathrm{d}\theta}{\mathrm{d}t} = 9$ and $t = 0$:

$9 = \mathrm{e}^{0}(4\sin 0 + 4B\cos 0) - 4\mathrm{e}^{0}(-\cos 0 + B\sin 0) - 4\sin 0 + 8\cos 0$

$\Rightarrow 9 = 4B + 4 + 8$

$\Rightarrow B = -\frac{3}{4}$

$\therefore \theta = \mathrm{e}^{-4t}\left(-\cos 4t - \frac{3}{4}\sin 4t\right) + \cos 4t + 2\sin 4t$

15) c)

As $t \to \infty$, $\mathrm{e}^{-4t} \to 0$

So $\theta \to \cos 4t + 2\sin 4t$

So as t becomes large, the pendulum approaches simple harmonic motion.

16)

$\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} + 6\frac{\mathrm{d}\theta}{\mathrm{d}t} + 9\theta = 9\mathrm{e}^{-t}$

Homogeneous Equation:

$\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} + 6\frac{\mathrm{d}\theta}{\mathrm{d}t} + 9\theta = 0$

Auxiliary Equation:

$m^2 + 6m + 9 = 0$

$\Rightarrow (m+3)^2 = 0$

$\Rightarrow m = -3$

Complementary Function:

$\theta = At\mathrm{e}^{-3t} + B\mathrm{e}^{-3t}$

Try this Particular Integral:

$\theta = p\mathrm{e}^{-t}$

$\frac{\mathrm{d}\theta}{\mathrm{d}t} = -p\mathrm{e}^{-t}$

$\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} = p\mathrm{e}^{-t}$

$\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} + 6\frac{\mathrm{d}\theta}{\mathrm{d}t} + 9\theta = 9\mathrm{e}^{-t}$

$\Rightarrow (p\mathrm{e}^{-t}) + 6(-p\mathrm{e}^{-t}) + 9(p\mathrm{e}^{-t}) = 9\mathrm{e}^{-t}$

$\Rightarrow (p - 6p + 9p)\mathrm{e}^{-t} = 9\mathrm{e}^{-t}$

$\Rightarrow 4p\mathrm{e}^{-t} = 9\mathrm{e}^{-t}$

$\Rightarrow 4p = 9$

$\Rightarrow p = \frac{9}{4}$

Particular Integral:

$\theta = \frac{9}{4}\mathrm{e}^{-t}$

General Solution:

$\theta = At\mathrm{e}^{-3t} + B\mathrm{e}^{-3t} + \frac{9}{4}\mathrm{e}^{-t}$

Substitute $\theta = 0$ and $t = 0$:

$0 = A \times 0 \times \mathrm{e}^0 + B\mathrm{e}^0 + \frac{9}{4}\mathrm{e}^0$

$\Rightarrow 0 = B + \frac{9}{4}$

$\Rightarrow B = -\frac{9}{4}$

$\therefore \theta = At\mathrm{e}^{-3t} - \frac{9}{4}\mathrm{e}^{-3t} + \frac{9}{4}\mathrm{e}^{-t}$

$\frac{\mathrm{d}\theta}{\mathrm{d}t} = -3At\mathrm{e}^{-3t} + A\mathrm{e}^{-3t} + \frac{27}{4}\mathrm{e}^{-3t} - \frac{9}{4}\mathrm{e}^{-t}$

Substitute $\frac{\mathrm{d}\theta}{\mathrm{d}t} = 0$ and $t = 0$:

$0 = -3A \times 0 \times \mathrm{e}^0 + A\mathrm{e}^0 + \frac{27}{4}\mathrm{e}^0 - \frac{9}{4}\mathrm{e}^0$

$\Rightarrow 0 = A + \frac{27}{4} - \frac{9}{4}$

$\Rightarrow A = -\frac{9}{2}$

$\therefore \theta = -\frac{9}{2}t\mathrm{e}^{-3t} - \frac{9}{4}\mathrm{e}^{-3t} + \frac{9}{4}\mathrm{e}^{-t}$

The pendulum is at the downward vertical when $\theta = 0$

$0 = -\frac{9}{2}t\mathrm{e}^{-3t} - \frac{9}{4}\mathrm{e}^{-3t} + \frac{9}{4}\mathrm{e}^{-t}$

$\Rightarrow 0 = \left(-\frac{1}{2}t - \frac{9}{4} + \frac{9}{4}\mathrm{e}^{2t}\right)\mathrm{e}^{-3t}$

$\Rightarrow \ 0 = (2t + 1 - \mathrm{e}^{2t})\mathrm{e}^{-3t}$

$\mathrm{e}^{-3t} \neq 0$

$\therefore 2t + 1 - \mathrm{e}^{2t} = 0$

$\Rightarrow \mathrm{e}^{2t} - 2t - 1 = 0$

Consider $\theta = \mathrm{e}^{2t} \Rightarrow \frac{\mathrm{d}\theta}{\mathrm{d}t} = 2\mathrm{e}^{2t}$, so $\left.\frac{\mathrm{d}\theta}{\mathrm{d}t}\right|_{t=0} = 2$

The tangent to $\theta = 2\mathrm{e}^{2t}$ that passes through $(0, 1)$ is

$\theta - 1 = 2(t - 0) \Rightarrow \theta = 2t + 1$

So $\mathrm{e}^{2t} > 2t + 1$, and so $\mathrm{e}^{2t} - 2t - 1 > 0$ for all $t > 0$

Hence $\theta > 0$ for all $t > 0$, and so as θ never becomes zero again, the pendulum never returns to the downward vertical.

17) a)

Take the positive direction to be the direction in which the spring is being compressed.

Spring Force $= -M\alpha^2 y$ N

So:

Resultant Force $F = ma$

$a = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}^2 y}{\mathrm{d}t^2}$

$\Rightarrow -M\alpha^2 y = M\frac{\mathrm{d}^2 y}{\mathrm{d}t^2}$

$\Rightarrow M\frac{\mathrm{d}^2 y}{\mathrm{d}t^2} + M\alpha^2 y = 0$

$\Rightarrow \frac{\mathrm{d}^2 y}{\mathrm{d}t^2} + \alpha^2 y = 0$

17) b)

At the instant the door first touches the spring, the spring has not yet been compressed, so $y = 0$ when $t = 0$.

The door moves into the spring with speed W, and we chose the direction of compression to be positive, so $\frac{\mathrm{d}y}{\mathrm{d}t} = W$ when $t = 0$.

17) c)

$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + \alpha^2 y = 0$

Auxiliary Equation:

$m^2 + \alpha^2 = 0$

$\Rightarrow m = \pm\alpha\mathrm{i}$

General solution:

$y = A\cos\alpha t + B\sin\alpha t$

Substitute $y = 0$ and $t = 0$:

$0 = A\cos 0 + B\sin 0 \Rightarrow A = 0$

$\therefore y = B\sin\alpha t$

$\frac{\mathrm{d}y}{\mathrm{d}t} = B\alpha\cos\alpha t$

Substitute $\frac{\mathrm{d}y}{\mathrm{d}t} = W$ and $t = 0$:

$W = B\alpha\cos 0 \Rightarrow B = \frac{W}{\alpha}$

$\therefore y = \frac{W}{\alpha}\sin\alpha t$

17) d)

$y = \frac{W}{\alpha}\sin\alpha t$

The maximum value of $\sin\alpha t$ is 1, so the maximum compression is $\frac{W}{\alpha}$

17) e)

Take the positive direction to be the direction in which the spring is being compressed.

Spring Force $= -M\alpha^2 y$ N

The resistive force opposes the motion:

Resistance Force $= -\lambda M v = -\lambda M \frac{\mathrm{d}y}{\mathrm{d}t}$

So:

Resultant Force $F = ma$

$a = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}^2y}{\mathrm{d}t^2}$

$\Rightarrow -M\alpha^2 y - \lambda M \frac{\mathrm{d}y}{\mathrm{d}t} = M \frac{\mathrm{d}^2y}{\mathrm{d}t^2}$

$\Rightarrow M \frac{\mathrm{d}^2y}{\mathrm{d}t^2} + \lambda M \frac{\mathrm{d}y}{\mathrm{d}t} + M\alpha^2 y = 0$

$\Rightarrow \frac{\mathrm{d}^2y}{\mathrm{d}t^2} + \lambda \frac{\mathrm{d}y}{\mathrm{d}t} + \alpha^2 y = 0$

17) f)

When $\lambda = 3\alpha$:

$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 3\alpha \frac{\mathrm{d}y}{\mathrm{d}t} + \alpha^2 y = 0$

Auxiliary Equation:

$m^2 + 3\alpha m + \alpha^2 = 0$

$b^2 - 4ac = (3\alpha)^2 - 4 \times 1 \times \alpha^2$

$= 9\alpha^2 - 4\alpha^2$

$= 5\alpha^2$

As α is a positive real constant, $5\alpha^2 > 0$

So the auxiliary equation has two distinct real roots, and hence the motion is Heavy Damping (overdamped).

17) g)

The door compresses the spring and is slowed by damping. It does not oscillate, but approaches equilibrium without repeated overshoot.

18) a)

$\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} + 4\frac{\mathrm{d}\theta}{\mathrm{d}t} + 13\theta = 26$

Homogeneous Equation:

$\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} + 4\frac{\mathrm{d}\theta}{\mathrm{d}t} + 13\theta = 0$

Auxiliary Equation:

$m^2 + 4m + 13 = 0$

$\Rightarrow (m+2)^2 - 4 + 13 = 0$

$\Rightarrow (m+2)^2 = -9$

$\Rightarrow m = -2 \pm 3\mathrm{i}$

Complementary Function:

$\theta = \mathrm{e}^{-2t}(A\cos 3t + B\sin 3t)$

Try this Particular Integral:

$\theta = k$

$\frac{\mathrm{d}\theta}{\mathrm{d}t} = 0$

$\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} = 0$

$\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} + 4\frac{\mathrm{d}\theta}{\mathrm{d}t} + 13\theta = 26$

$\Rightarrow (0) + 4(0) + 13(k) = 26$

$\Rightarrow 13k = 26$

$\Rightarrow k = 2$

Particular Integral:

$\theta = 2$

General Solution:

$\theta = \mathrm{e}^{-2t}(A\cos 3t + B\sin 3t) + 2$

Substitute $\theta = 3$ and $t = 0$:

$3 = \mathrm{e}^0(A\cos 0 + B\sin 0) + 2$

$\Rightarrow 3 = A + 2$

$\Rightarrow A = 1$

$\therefore \theta = \mathrm{e}^{-2t}(\cos 3t + B\sin 3t) + 2$

$\frac{\mathrm{d}\theta}{\mathrm{d}t} = \mathrm{e}^{-2t}(-3\sin 3t + 3B\cos 3t) - 2\mathrm{e}^{-2t}(\cos 3t + B\sin 3t)$

Substitute $\frac{\mathrm{d}\theta}{\mathrm{d}t} = 0$ and $t = 0$:

$0 = \mathrm{e}^0(-3\sin 0 + 3B\cos 0) - 2\mathrm{e}^0(\cos 0 + B\sin 0)$

$\Rightarrow 0 = 3B - 2$

$\Rightarrow B = \frac{2}{3}$

Particular Solution:

$\therefore \theta = \mathrm{e}^{-2t}\left(\cos 3t + \frac{2}{3}\sin 3t\right) + 2$

18) b)

$\theta = 2 + \mathrm{e}^{-2t}\left(\cos 3t + \frac{2}{3}\sin 3t\right)$

$\cos 3t + \frac{2}{3}\sin 3t \equiv R\cos(3t - \alpha)$

$\equiv R(\cos 3t \cos\alpha + \sin 3t \sin\alpha)$

$\equiv R\cos\alpha \cos 3t + R\sin\alpha \sin 3t$

$R\cos\alpha = 1 \Rightarrow \cos\alpha = \frac{1}{R}$

$R\sin\alpha = \frac{2}{3} \Rightarrow \sin\alpha = \frac{2}{3R}$

$R = \sqrt{1^2 + \left(\frac{2}{3}\right)^2} = \sqrt{\frac{13}{9}} = \frac{\sqrt{13}}{3}$

$\alpha = \arctan\frac{2}{3} = 0.588$ (3 SF)

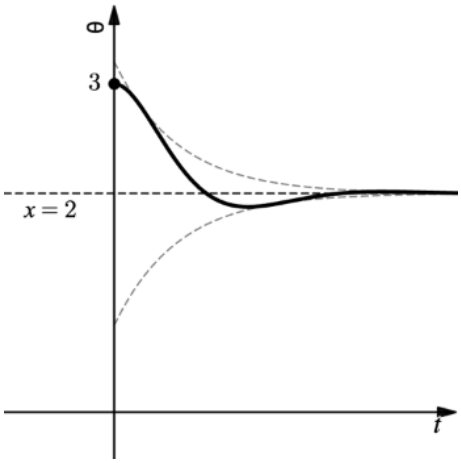
$\therefore \theta = 2 + \frac{\sqrt{13}}{3}\mathrm{e}^{-2t}\cos\left(3t - \arctan\frac{2}{3}\right)$

18) c)

As $t \to \infty$, $\mathrm{e}^{-2t} \to 0$ so $\theta \to 2$

So the oscillations die away because of the damping, and the door settles to a constant angle of 2 radians from the vertical.

18) d)



19) a)

Taking downwards as positive, the weight acts downwards and hence is positive:

Weight $= Mg$ N

Tension will act opposite to the weight, and is therefore negative:

Tension $= -Cy$ N

The resistive force opposes the motion:

Resistive Force $= -Du = -D\frac{\mathrm{d}y}{\mathrm{d}t}$

So:

Resultant Force $F = ma$

$a = \frac{\mathrm{d}u}{\mathrm{d}t} = \frac{\mathrm{d}^2y}{\mathrm{d}t^2}$

$\Rightarrow Mg - Cy - D\frac{\mathrm{d}y}{\mathrm{d}t} = M\frac{\mathrm{d}^2y}{\mathrm{d}t^2}$

$\Rightarrow M\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + D\frac{\mathrm{d}y}{\mathrm{d}t} + Cy = Mg$

19) b)

Homogeneous Equation:

$M\frac{\mathrm{d}^2y}{\mathrm{d}t^2}+D\frac{\mathrm{d}y}{\mathrm{d}t}+Cy=0$

Auxiliary Equation:

$Mr^2+Dr+C=0$

$\Rightarrow M\left(r^2+\frac{D}{M}r\right)+C=0$

$\Rightarrow M\left(\left(r+\frac{D}{2M}\right)^2-\frac{D^2}{4M^2}\right)+C=0$

$\Rightarrow M\left(r+\frac{D}{2M}\right)^2-\frac{D^2}{4M}+C=0$

$\Rightarrow \left(r+\frac{D}{2M}\right)^2=\frac{D^2}{4M^2}-\frac{C}{M}$

$\Rightarrow \left(r+\frac{D}{2M}\right)^2=\frac{D^2-4MC}{4M^2}$

$\Rightarrow r=-\frac{D}{2M}\pm\frac{\sqrt{D^2-4MC}}{2M}$

As $4CM>D^2$ we can write:

$r=-\frac{D}{2M}\pm\frac{\sqrt{-(4MC-D^2)}}{2M}$

$\Rightarrow r=-\frac{D}{2M}\pm \mathrm{i}\frac{\sqrt{4MC-D^2}}{2M}$

Complementary Function:

$y=\mathrm{e}^{-\frac{D}{2M}t}\left(P\cos\left(\frac{\sqrt{4MC-D^2}}{2M}t\right)+Q\sin\left(\frac{\sqrt{4MC-D^2}}{2M}t\right)\right)$

Try this Particular Integral:

$y=k$

$\frac{\mathrm{d}y}{\mathrm{d}t}=0$

$\frac{\mathrm{d}^2y}{\mathrm{d}t^2}=0$

$M\frac{\mathrm{d}^2y}{\mathrm{d}t^2}+D\frac{\mathrm{d}y}{\mathrm{d}t}+Cy=M\mathrm{g}$

$\Rightarrow M(0)+D(0)+C(k)=M\mathrm{g}$

$\Rightarrow k=\frac{M\mathrm{g}}{C}$

Particular Integral:

$y=\frac{M\mathrm{g}}{C}$

General Solution:

$y=\mathrm{e}^{-\frac{D}{2M}t}\left(P\cos\left(\frac{\sqrt{4MC-D^2}}{2M}t\right)+Q\sin\left(\frac{\sqrt{4MC-D^2}}{2M}t\right)\right)+\frac{M\mathrm{g}}{C}$

19) c)

Substitute $C = 25$, $D = 24$, $M = 9$, $\mathrm{g} = 10$:

$$y = \mathrm{e}^{-\frac{24}{2\times9}t}\left(P\cos\left(\frac{\sqrt{4\times9\times25-24^2}}{2\times9}t\right) + Q\sin\left(\frac{\sqrt{4\times9\times25-24^2}}{2\times9}t\right)\right) + \frac{9\times10}{25}$$

$$y = \mathrm{e}^{-\frac{4}{3}t}(P\cos t + Q\sin t) + \frac{18}{5}$$

Substitute $y = 0$ and $t = 0$:

$$0 = \mathrm{e}^{0}(P\cos 0 + Q\sin 0) + \frac{18}{5}$$

$$\Rightarrow P = -\frac{18}{5}$$

$$y = \mathrm{e}^{-\frac{4}{3}t}\left(-\frac{18}{5}\cos t + Q\sin t\right) + \frac{18}{5}$$

$$\frac{\mathrm{d}y}{\mathrm{d}t} = \mathrm{e}^{-\frac{4}{3}t}\left(\frac{18}{5}\sin t + Q\cos t\right) - \frac{4}{3}\mathrm{e}^{-\frac{4}{3}t}\left(-\frac{18}{5}\cos t + Q\sin t\right)$$

Substitute $\frac{\mathrm{d}y}{\mathrm{d}t} = 9$ and $t = 0$:

$$9 = \mathrm{e}^{0}\left(\frac{18}{5}\sin 0 + Q\cos 0\right) - \frac{4}{3}\mathrm{e}^{0}\left(-\frac{18}{5}\cos 0 + Q\sin 0\right)$$

$$\Rightarrow 9 = Q + \frac{24}{5}$$

$$\Rightarrow Q = \frac{21}{5}$$

$$\therefore y = \mathrm{e}^{-\frac{4}{3}t}\left(-\frac{18}{5}\cos t + \frac{21}{5}\sin t\right) + \frac{18}{5}$$

19) d)

$$y = \mathrm{e}^{-\frac{4}{3}t}\left(-\frac{18}{5}\cos t + \frac{21}{5}\sin t\right) + \frac{18}{5}$$

The crate reaches its greatest extension when $\frac{\mathrm{d}y}{\mathrm{d}t} = 0$

$$\frac{\mathrm{d}y}{\mathrm{d}t} = \mathrm{e}^{-\frac{4}{3}t}\left(\frac{18}{5}\sin t + \frac{21}{5}\cos t\right) - \frac{4}{3}\mathrm{e}^{-\frac{4}{3}t}\left(-\frac{18}{5}\cos t + \frac{21}{5}\sin t\right)$$

$$\frac{\mathrm{d}y}{\mathrm{d}t} = 0 \Rightarrow \mathrm{e}^{-\frac{4}{3}t}\left(\frac{18}{5}\sin t + \frac{21}{5}\cos t\right) - \frac{4}{3}\mathrm{e}^{-\frac{4}{3}t}\left(-\frac{18}{5}\cos t + \frac{21}{5}\sin t\right) = 0$$

$$\Rightarrow \left(\frac{18}{5}\sin t + \frac{21}{5}\cos t\right) - \frac{4}{3}\left(-\frac{18}{5}\cos t + \frac{21}{5}\sin t\right) = 0$$

$$\Rightarrow \frac{18}{5}\sin t + \frac{21}{5}\cos t + \frac{24}{5}\cos t - \frac{28}{5}\sin t = 0$$

$$\Rightarrow 9\cos t - 2\sin t = 0$$

$$\Rightarrow \frac{9}{2} = \frac{\sin t}{\cos t}$$

$$\Rightarrow \tan t = \frac{9}{2}$$

$$\Rightarrow t = 1.35 \text{ s (3 SF)}$$

19) e)

As $t \to \infty$, $\mathrm{e}^{-\frac{4}{3}t} \to 0$ so $y \to \frac{18}{5}$

So the equilibrium extension of the cable is 3.6 m

19) f)

With no resistive force, we have:

$\Rightarrow Mg - Cy = M\frac{\mathrm{d}^2y}{\mathrm{d}t^2}$

$\Rightarrow M\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + Cy = Mg$

Substitute $C = 25$, $M = 9$, $g = 10$:

$9\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 25y = 90$

Homogeneous Equation:

$9\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 25y = 0$

Auxiliary Equation:

$9r^2 + 25 = 0$

$\Rightarrow r^2 = -\frac{25}{9}$

$\Rightarrow r = \pm\frac{5}{3}\mathrm{i}$

Complementary Function:

$y = P\cos\frac{5}{3}t + Q\sin\frac{5}{3}t$

Try this Particular Integral:

$y = k$

$\frac{\mathrm{d}y}{\mathrm{d}t} = 0$

$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} = 0$

$9\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 25y = 90$

$\Rightarrow 9(0) + 25(k) = 90$

$\Rightarrow k = \frac{18}{5}$

Particular Integral:

$y = \frac{18}{5}$

General Solution:

$y = P\cos\frac{5}{3}t + Q\sin\frac{5}{3}t + \frac{18}{5}$

Substitute $y = 0$ and $t = 0$:

$0 = P\cos 0 + Q\sin 0 + \frac{18}{5}$

$\Rightarrow P = -\frac{18}{5}$

$y = -\frac{18}{5}\cos\frac{5}{3}t + Q\sin\frac{5}{3}t + \frac{18}{5}$

$\frac{dy}{dt} = 6\sin\frac{5}{3}t + \frac{5}{3}Q\cos\frac{5}{3}t$

Substitute $\frac{dy}{dt} = 9$ and $t = 0$:

$9 = 6\sin 0 + \frac{5}{3}Q\cos 0$

$\Rightarrow \frac{5}{3}Q = 9$

$\Rightarrow Q = \frac{27}{5}$

$\therefore y = -\frac{18}{5}\cos\frac{5}{3}t + \frac{27}{5}\sin\frac{5}{3}t + \frac{18}{5}$

19) g)

The damped model was:

$y = \mathrm{e}^{-\frac{4}{3}t}\left(-\frac{18}{5}\cos t + \frac{21}{5}\sin t\right) + \frac{18}{5}$

From part d), we found that the maximum extension occurs when $\tan t = \frac{9}{2}$

$\Rightarrow t = \arctan\frac{9}{2}\ (= 1.352\ldots)$

$\therefore y|_{t=\arctan\frac{9}{2}} = \mathrm{e}^{-\frac{4}{3}\arctan\frac{9}{2}}\left(-\frac{18}{5}\cos\left(\arctan\frac{9}{2}\right) + \frac{21}{5}\sin\left(\arctan\frac{9}{2}\right)\right) + \frac{18}{5}$

$= 4.147078965\ldots$

$= 4.15$ (3 SF)

The undamped model was:

$y = -\frac{18}{5}\cos\frac{5}{3}t + \frac{27}{5}\sin\frac{5}{3}t\ + \frac{18}{5}$

$\frac{\mathrm{d}y}{\mathrm{d}t} = 6\sin\frac{5}{3}t + 9\cos\frac{5}{3}t$

Maximum extension occurs when $\frac{\mathrm{d}y}{\mathrm{d}t} = 0$

$\Rightarrow 6\sin\frac{5}{3}t + 9\cos\frac{5}{3}t = 0$

$\Rightarrow \frac{\sin\frac{5}{3}t}{\cos\frac{5}{3}t} = -\frac{3}{2}$

$\Rightarrow \tan\frac{5}{3}t = -\frac{3}{2}$

$\Rightarrow t = \frac{3}{5}\left(\arctan\left(-\frac{3}{2}\right) + \pi\right)\ (= 1.295\ldots)$

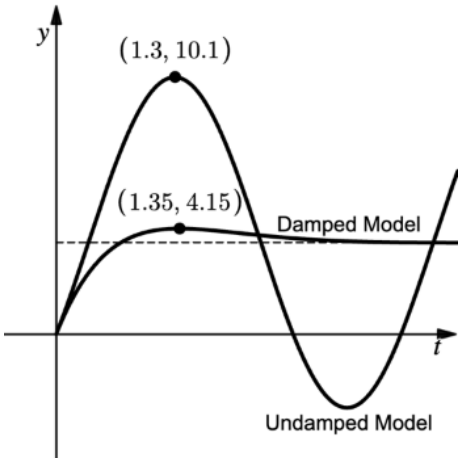
$\therefore y|_{t=\frac{3}{5}\left(\arctan\left(-\frac{3}{2}\right)+\pi\right)} = -\frac{18}{5}\cos\left(\frac{5}{3}\times\frac{3}{5}\left(\arctan\left(-\frac{3}{2}\right) + \pi\right)\right) + \frac{27}{5}\sin\left(\frac{5}{3}\times\frac{3}{5}\left(\arctan\left(-\frac{3}{2}\right) + \pi\right)\right) + \frac{18}{5}$

$= 10.0899923\ldots$

$= 10.1$ (3 SF)

$10.1 > 4.15$ so the greatest extension predicted by the undamped model is larger than the greatest extension predicted by the damped model.

19) h)



20) a)

$\ddot{x} + \lambda\dot{x} = 0$

$v = \dot{x} = \frac{\mathrm{d}x}{\mathrm{d}t}$

So $\ddot{x} = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}v}{\mathrm{d}x} \times \frac{\mathrm{d}x}{\mathrm{d}t} = v\frac{\mathrm{d}v}{\mathrm{d}x}$

$\therefore v\frac{\mathrm{d}v}{\mathrm{d}x} + \lambda v = 0$

20) b)

$v\frac{\mathrm{d}v}{\mathrm{d}x} + \lambda v = 0$

As the particle is moving, $v \neq 0$ so we can divide through by v:

$\Rightarrow \frac{\mathrm{d}v}{\mathrm{d}x} + \lambda = 0$

$\Rightarrow \frac{\mathrm{d}v}{\mathrm{d}x} = -\lambda$

$\Rightarrow v = -\lambda x + c$

Initially $x = 0$ and $v = u$:

$u = -\lambda \times 0 + c$

$\Rightarrow c = u$

$\therefore v = -\lambda x + u$

$\Rightarrow v = u - \lambda x$

20) c)

$v = u - \lambda x$

At the point A, the particle comes to instantaneous rest, so $v = 0$ when $x = d$

$0 = u - \lambda d$

$\Rightarrow u = \lambda d$

20) d)

$v = u - \lambda x$

$\Rightarrow v = \lambda d - \lambda x$

$\Rightarrow v = \lambda(d - x)$

20) e)

$v = \lambda(d - x)$

$z = \frac{1}{v} = \frac{1}{\lambda(d-x)}$

So the mean value of z with respect to displacement from $x = 0$ to $x = \frac{1}{2}d$ is

$\frac{1}{\frac{1}{2}d-0}\int_0^{\frac{1}{2}d} \frac{1}{\lambda(d-x)}\,\mathrm{d}x$

$= \frac{2}{\lambda d}\int_0^{\frac{1}{2}d} \frac{1}{d-x}\,\mathrm{d}x$

$= \frac{2}{\lambda d}\left[-\ln|d-x|\right]_0^{\frac{1}{2}d}$

$= \frac{2}{\lambda d}\left(-\ln\left|d-\frac{1}{2}d\right| - (-\ln|d-0|)\right)$

$= \frac{2}{\lambda d}\left(-\ln\frac{1}{2}d + \ln d\right)$

$= \frac{2}{\lambda d}\left(\ln d - \ln\frac{1}{2}d\right)$

$= \frac{2}{\lambda d}\ln\left(\frac{d}{\frac{1}{2}d}\right)$

$= \frac{2\ln 2}{\lambda d}$

$= \frac{\ln 4}{u}$

21) a)

The amplitude at time t is the coefficient of $\cos 3t$, so:

$5\mathrm{e}^{-0.2t}$

21) b)

The period is $T = \frac{2\pi}{3}$

Hence the ratio of the amplitudes at times t and $t + T$ is:

$\frac{5e^{-0.2(t+T)}}{5e^{-0.2t}} = e^{-0.2T}$

$= e^{-0.2\times\frac{2\pi}{3}}$

$= e^{-\frac{2\pi}{15}}$

22) a)

$\frac{d^2y}{dt^2} + 2c\frac{dy}{dt} + 10y = 0$

Auxiliary Equation:

$m^2 + 2cm + 10 = 0$

$\Rightarrow (m + c)^2 - c^2 + 10 = 0$

$\Rightarrow (m + c)^2 = c^2 - 10$

$\Rightarrow m = -c \pm \sqrt{c^2 - 10}$

The motion is oscillatory, so we require $c^2 < 10$

$\therefore m = -c \pm i\sqrt{10 - c^2}$

General Solution:

$y = e^{-ct}(A\cos\omega t + B\sin\omega t)$

where $\omega = \sqrt{10 - c^2}$

22) b)

One oscillation takes time T, where:

$T = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{10-c^2}}$

After one oscillation, the amplitude is multiplied by e^{-cT}

We are told the amplitude after each complete oscillation is 0.9 times the previous amplitude, so:

$e^{-cT} = 0.9$

22) c)

$e^{-cT} = 0.9$

$\Rightarrow e^{-c\left(\frac{2\pi}{\sqrt{10-c^2}}\right)} = 0.9$

$\Rightarrow -c\left(\frac{2\pi}{\sqrt{10-c^2}}\right) = \ln 0.9$

$\Rightarrow \frac{4\pi^2c^2}{10-c^2} = (\ln 0.9)^2$

$\Rightarrow 4\pi^2c^2 = 10(\ln 0.9)^2 - (\ln 0.9)^2c^2$

$\Rightarrow 4\pi^2c^2 + (\ln 0.9)^2c^2 = 10(\ln 0.9)^2$

$\Rightarrow (4\pi^2 + (\ln 0.9)^2)c^2 = 10(\ln 0.9)^2$

$\Rightarrow c^2 = \frac{10(\ln 0.9)^2}{4\pi^2+(\ln 0.9)^2}$

$\Rightarrow c = \sqrt{\frac{10(\ln 0.9)^2}{4\pi^2+(\ln 0.9)^2}}$

$= 0.0530196636\ldots$

$= 0.0530$ (3 SF)

22) d)

$y = e^{-ct}(A\cos\omega t + B\sin\omega t)$

where $\omega = \sqrt{10 - c^2}$

Substitute $y = 0$ and $t = 0$:

$0 = e^0(A\cos 0 + B\sin 0) \Rightarrow A = 0$

$\therefore y = Be^{-ct}\sin\omega t$

$\frac{dy}{dt} = B\omega e^{-ct}\cos\omega t - Bce^{-ct}\sin\omega t$

Substitute $\frac{dy}{dt} = 15$ and $t = 0$:

$15 = B\omega e^0\cos 0 - Bce^0\sin 0$

$\Rightarrow 15 = B\omega$

$\Rightarrow B = \frac{15}{\omega}$

$\therefore y = \frac{15}{\omega}e^{-ct}\sin\omega t$

$c = \sqrt{\frac{10(\ln 0.9)^2}{4\pi^2+(\ln 0.9)^2}} = 0.0530196636\ldots$

So $\omega = \sqrt{10 - 0.05301\ldots^2} = 3.16183315\ldots$

$B = \frac{15}{\omega} = \frac{15}{3.1618\ldots} = 4.74408333\ldots$

So

$y = 4.74e^{-0.0530t}\sin 3.16t$

23) a)

Take to the right as positive.

The restoring force acts towards E, so in the negative x-direction:

Restoring Force $= -4mx$ N

So:

Resultant Force $F = ma$

$a = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$\Rightarrow -4mx = m\frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$\Rightarrow m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 4mx = 0$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 4x = 0$

23) b)

Auxiliary Equation:

$m^2 + 4 = 0$

$\Rightarrow m^2 = -4$

$\Rightarrow m = \pm 2\mathrm{i}$

General Solution:

$x = A\cos 2t + B\sin 2t$

Substitute $x = 1$ and $t = 0$:

$1 = A\cos 0 + B\sin 0$

$\Rightarrow A = 1$

$\therefore x = \cos 2t + B\sin 2t$

$\frac{\mathrm{d}x}{\mathrm{d}t} = -2\sin 2t + 2B\cos 2t$

Substitute $\frac{\mathrm{d}x}{\mathrm{d}t} = 2$ and $t = 0$:

$2 = -2\sin 0 + 2B\cos 0$

$\Rightarrow 2B = 2$

$\Rightarrow B = 1$

$\therefore x = \cos 2t + \sin 2t$

23) c)

Amplitude $= \sqrt{1^2 + 1^2} = \sqrt{2}$

Period $= \frac{2\pi}{2} = \pi$

23) d)

Simple Harmonic Motion

23) e)

The resistive force opposes the direction of motion.

Resistive Force $= -2mv$ N

$v = \frac{\mathrm{d}x}{\mathrm{d}t}$

So:

Resultant Force $F = ma$

$\Rightarrow -4mx - 2m\frac{\mathrm{d}x}{\mathrm{d}t} = m\frac{\mathrm{d}^2x}{\mathrm{d}t^2}$

$\Rightarrow m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 2m\frac{\mathrm{d}x}{\mathrm{d}t} + 4mx = 0$

$\Rightarrow \frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 2\frac{\mathrm{d}x}{\mathrm{d}t} + 4x = 0$

23) f)

Auxiliary Equation:

$m^2 + 2m + 4 = 0$

$\Rightarrow (m+1)^2 - 1 + 4 = 0$

$\Rightarrow (m+1)^2 = -3$

$\Rightarrow m = -1 \pm \sqrt{3}\mathrm{i}$

General Solution:

$x = \mathrm{e}^{-t}\left(A\cos\sqrt{3}t + B\sin\sqrt{3}t\right)$

23) g)

The roots of the auxiliary equation are complex, and so oscillations occur but decay with time. This is light damping (underdamping).

23) h)

Try this Particular Integral:

$x = p\cos t + q\sin t$

$\frac{\mathrm{d}x}{\mathrm{d}t} = -p\sin t + q\cos t$

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -p\cos t - q\sin t$

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 2\frac{\mathrm{d}x}{\mathrm{d}t} + 4x = 4\cos t$

$\Rightarrow (-p\cos t - q\sin t) + 2(-p\sin t + q\cos t) + 4(p\cos t + q\sin t) = 4\cos t$

$\Rightarrow (-p + 2q + 4p)\cos t + (-q - 2p + 4q)\sin t = 4\cos t$

$\Rightarrow (3p + 2q)\cos t + (-2p + 3q)\sin t = 4\cos t$

Compare Coefficients:

$3p + 2q = 4$ (1)

$-2p + 3q = 0$ (2)

Solve (1) & (2):

$p = \frac{12}{13}$ and $q = \frac{8}{13}$

Particular Integral:

$x = \frac{12}{13}\cos t + \frac{8}{13}\sin t$

General Solution:

$x = \mathrm{e}^{-t}\left(A\cos\sqrt{3}t + B\sin\sqrt{3}t\right) + \frac{12}{13}\cos t + \frac{8}{13}\sin t$

Substitute $x = 1$ and $t = 0$:

$1 = \mathrm{e}^0(A\cos 0 + B\sin 0) + \frac{12}{13}\cos 0 + \frac{8}{13}\sin 0$

$\Rightarrow 1 = A + \frac{12}{13}$

$\Rightarrow A = \frac{1}{13}$

$\therefore x = \mathrm{e}^{-t}\left(\frac{1}{13}\cos\sqrt{3}t + B\sin\sqrt{3}t\right) + \frac{12}{13}\cos t + \frac{8}{13}\sin t$

$\frac{\mathrm{d}x}{\mathrm{d}t} = \mathrm{e}^{-t}\left(-\frac{\sqrt{3}}{13}\sin\sqrt{3}t + B\sqrt{3}\cos\sqrt{3}t\right) - \mathrm{e}^{-t}\left(\frac{1}{13}\cos\sqrt{3}t + B\sin\sqrt{3}t\right) - \frac{12}{13}\sin t + \frac{8}{13}\cos t$

Substitute $\frac{\mathrm{d}x}{\mathrm{d}t} = 2$ and $t = 0$:

$2 = \mathrm{e}^0\left(-\frac{\sqrt{3}}{13}\sin 0 + B\sqrt{3}\cos 0\right) - \mathrm{e}^0\left(\frac{1}{13}\cos 0 + B\sin 0\right) - \frac{12}{13}\sin 0 + \frac{8}{13}\cos 0$

$\Rightarrow 2 = B\sqrt{3} - \frac{1}{13} + \frac{8}{13}$

$\Rightarrow B\sqrt{3} = \frac{19}{13}$

$\Rightarrow B = \frac{19}{13\sqrt{3}} = \frac{19\sqrt{3}}{39}$

$\therefore x = \mathrm{e}^{-t}\left(\frac{1}{13}\cos\sqrt{3}t + \frac{19\sqrt{3}}{39}\sin\sqrt{3}t\right) + \frac{12}{13}\cos t + \frac{8}{13}\sin t$

23) i)

As $t \to \infty$, $e^{-t} \to 0$

$x \to \frac{12}{13}\cos t + \frac{8}{13}\sin t$

Period $= 2\pi = 6.28318 \ldots$ seconds

So the period is just over 6 seconds

23) j)

Amplitude $= \sqrt{\left(\frac{12}{13}\right)^2 + \left(\frac{8}{13}\right)^2} = \frac{4\sqrt{13}}{13}$

23) k)

$\frac{1}{13}\cos\sqrt{3}t + \frac{19\sqrt{3}}{39}\sin\sqrt{3}t \equiv C\cos(\sqrt{3}t - \beta)$

$\Rightarrow \frac{1}{13}\cos\sqrt{3}t + \frac{19\sqrt{3}}{39}\sin\sqrt{3}t \equiv C(\cos\sqrt{3}t\cos\beta + \sin\sqrt{3}t\sin\beta)$

$\Rightarrow \frac{1}{13}\cos\sqrt{3}t + \frac{19\sqrt{3}}{39}\sin\sqrt{3}t \equiv C\cos\beta\cos\sqrt{3}t + C\sin\beta\sin\sqrt{3}t$

So:

$\frac{1}{13} = C\cos\beta \Rightarrow \cos\beta = \frac{1}{13C}$

$\frac{19\sqrt{3}}{39} = C\sin\beta \Rightarrow \sin\beta = \frac{19\sqrt{3}}{39C}$

$\left(\frac{1}{13C}\right)^2 + \left(\frac{19\sqrt{3}}{39C}\right)^2 = 1$

$\Rightarrow \frac{1}{169C^2} + \frac{361}{507C^2} = 1$

$\Rightarrow \frac{1}{169} + \frac{361}{507} = C^2$

$\Rightarrow C^2 = \frac{28}{39}$

$\Rightarrow C = \frac{2\sqrt{273}}{39}$

$\tan\beta = \frac{\sin\beta}{\cos\beta} = \frac{\frac{19\sqrt{3}}{39C}}{\frac{1}{13C}} = \frac{19\sqrt{3}}{3}$

$\Rightarrow \beta = \arctan\left(\frac{19\sqrt{3}}{3}\right) = 1.48$ (3 SF)

$\therefore \frac{1}{13}\cos\sqrt{3}t + \frac{19\sqrt{3}}{39}\sin\sqrt{3}t \equiv \frac{2\sqrt{273}}{39}\cos\left(\sqrt{3}t - \arctan\left(\frac{19\sqrt{3}}{3}\right)\right)$

23) l)

From part k), we have found that the solution may be written as

$x(t) = \mathrm{e}^{-t}\left(\frac{2\sqrt{273}}{39}\cos\left(\sqrt{3}t - \arctan\left(\frac{19\sqrt{3}}{3}\right)\right)\right) + \frac{12}{13}\cos t + \frac{8}{13}\sin t$

Let $x_S(t) = \frac{12}{13}\cos t + \frac{8}{13}\sin t$

So $x(t) = \mathrm{e}^{-t}\left(\frac{2\sqrt{273}}{39}\cos\left(\sqrt{3}t - \arctan\left(\frac{19\sqrt{3}}{3}\right)\right)\right) + x_S(t)$

Now, for all values of t, we have that

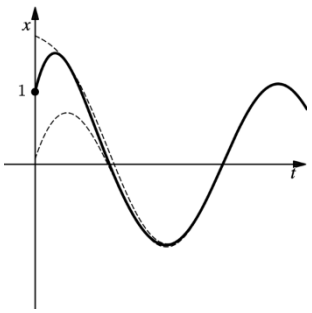
$-1 \leq \cos\left(\sqrt{3}t - \arctan\left(\frac{19\sqrt{3}}{3}\right)\right) \leq 1$

$\Rightarrow -\frac{2\sqrt{273}}{39}\mathrm{e}^{-t} \leq \frac{2\sqrt{273}}{39}\mathrm{e}^{-t}\cos\left(\sqrt{3}t - \arctan\left(\frac{19\sqrt{3}}{3}\right)\right) \leq \frac{2\sqrt{273}}{39}\mathrm{e}^{-t}$

$\Rightarrow x_S(t) - \frac{2\sqrt{273}}{39}\mathrm{e}^{-t} \leq \frac{2\sqrt{273}}{39}\mathrm{e}^{-t}\cos\left(\sqrt{3}t - \arctan\left(\frac{19\sqrt{3}}{3}\right)\right) + x_S(t) \leq x_S(t) + \frac{2\sqrt{273}}{39}\mathrm{e}^{-t}$

$\Rightarrow x_S(t) - \frac{2\sqrt{273}}{39}\mathrm{e}^{-t} \leq x(t) \leq x_S(t) + \frac{2\sqrt{273}}{39}\mathrm{e}^{-t}$ as required

23) m)



24) a)

$x = \mathrm{e}^{-\frac{t}{2}}(A\cos 2t + B\sin 2t) + \cos t - \sin t$

Consider $x = \mathrm{e}^{-\frac{t}{2}}(A\cos 2t + B\sin 2t)$ first

This corresponds to an auxiliary equation with roots $m = -\frac{1}{2} \pm 2\mathrm{i}$

$\Rightarrow m + \frac{1}{2} = \pm 2\mathrm{i}$

$\Rightarrow \left(m + \frac{1}{2}\right)^2 = -4$

$\Rightarrow m^2 + m + \frac{1}{4} + 4 = 0$

$\Rightarrow m^2 + m + \frac{17}{4} = 0$

So the complementary function satisfies the homogeneous equation

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \frac{\mathrm{d}x}{\mathrm{d}t} + \frac{17}{4}x = 0$

Now consider $x = \cos t - \sin t$

$\frac{\mathrm{d}x}{\mathrm{d}t} = -\sin t - \cos t$

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -\cos t + \sin t$

$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \frac{\mathrm{d}x}{\mathrm{d}t} + \frac{17}{4}x$

$= (-\cos t + \sin t) + (-\sin t - \cos t) + \frac{17}{4}(\cos t - \sin t)$

$= -\cos t + \sin t - \sin t - \cos t + \frac{17}{4}\cos t - \frac{17}{4}\sin t$

$= \frac{9}{4}\cos t - \frac{17}{4}\sin t$

$\therefore \frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \frac{\mathrm{d}x}{\mathrm{d}t} + \frac{17}{4}x = \frac{9}{4}\cos t - \frac{17}{4}\sin t$

$\Rightarrow 4\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + 4\frac{\mathrm{d}x}{\mathrm{d}t} + 17x = 9\cos t - 17\sin t$

24) b)

